

AP CSP | Lesson 07: Data Quality & Cleaning

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

12 original questions, 30 classroom marks, about 40 minutes. Q5 requires exactly two selections. Explain every cleaning decision; retain unusual values unless evidence justifies a change. Use the synthetic commute dataset and its stated rules.

1. Which item is metadata describing a commute table? [1 mark]

A. One response's time: 28. B. One response's mode: Bus.
C. One response's ID: 15. D. Time means one-way duration in minutes.

Answer: _____

2. A time is stored as 30 minutes. Only its unit metadata is changed to hours. What happens? [1 mark]

A. The number automatically becomes 0.5. B. The number stays 30; the label is now wrong.
C. It is correctly converted to 30 hours. D. The measurement is deleted.

Answer: _____

3. Valid mode labels are Walk, Bike and Bus. Which action preserves the meaning of the raw value "bus"? [1 mark]

A. Change it to Bike. B. Replace it with 0.
C. Trim spaces and standardize it to Bus. D. Delete all nonstandard labels.

Answer: _____

4. ID 09 has a missing time. The rule excludes missing times from this analysis and preserves raw data. What should happen? [1 mark]

A. Keep the raw row; exclude and log it. B. Replace the blank with 0.
C. Copy the previous row's time. D. Erase the raw row and its history.

Answer: _____

5. ID 03 has two identical records; ID 14 reports an unverified 120 minutes. Rules keep one identical copy and retain unusual values unless proven erroneous. Which TWO actions comply? [1 mark]

A. Remove both ID 03 records. B. Keep the first 03; log removal of the copy.
C. Delete ID 14 because 120 looks high. D. Keep ID 14 and flag it for review.

Answer: _____

6. Two tables describe the same responses in different row orders. Both use the same unique response IDs. How should they be combined? [1 mark]

A. Match by row position. B. Match similar numbers, ignoring IDs.
C. Match by the shared response ID. D. Treat times and distances as interchangeable.

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

A record is 30. Metadata: “one-way time in minutes; fictional voluntary survey, one school, one morning.”

- (a) Identify field metadata and explain its use.
- (b) Identify source metadata and explain its use.
- (c) If only the unit label becomes hours, does the number change?
- (d) Express the original duration correctly in hours.

8. [4 marks]

Both tables use the same unique response IDs from one survey.

A (ID, min): (02, 25), (03, 12). B (ID, km): (03, 0.8), (02, 4.0).

- (a) Give the two correctly combined records.
- (b) Why would matching by row position fail?
- (c) Give one metadata check needed when combining different sources.

9. [4 marks]

Elapsed one-way time: ID 09 is blank; ID 10 is -5.

Policy: preserve raw data; exclude and log missing or negative times.

- (a) Explain each record's issue.
- (b) State the policy's actions, including what to log.
- (c) Why are replacing the blank with 0 and -5 with +5 unjustified?

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Written responses | Questions 10-12

10. [4 marks]

Raw data: 16 rows; 09 missing time; 10 negative time; later 03 identical; 15 mode “ bus ”; 14 time 120, unverified.

Exclude missing/negative times and the extra copy; normalize labels; retain unusual values.

- (a) Calculate the retained row count.
- (b) State ID 15’s change and its effect on row count.
- (c) State ID 14’s treatment and justify it.

11. [4 marks]

Hypothetical: the same response ID has times 12 and 30. Only one time per response is intended.

- (a) Why is the identical-copy rule insufficient?
- (b) Give one investigation step.
- (c) If unresolved, give a transparent handling decision.
- (d) State two items to retain in the decision log.

12. [4 marks]

A voluntary single-school survey compares time by mode. It proposes collecting full names and exact addresses too.

- (a) Give a smaller collection plan.
- (b) Explain one privacy risk, including possible data linkage.
- (c) Would cleaning justify conclusions about all students? Explain.
- (d) Would more same-source volunteers guarantee no bias? Explain.
